1. Indefinite integrals

Definition. Let f be defined on an open interval I. If $F' = f$ on I, we write

$$\boxed{\int f(x)\,dx = F(x) + C, \qquad C \in \mathbb{R}.}$$

The *indefinite integral* denotes the family of all antiderivatives of f on I. Every member has derivative f; conversely, the Mean Value Theorem shows that any two antiderivatives on an interval differ by a constant. FTC I ensures that a continuous function on an interval has an antiderivative.

$\displaystyle\int_a^b f(x)\,dx$, fixed bounds	a number, when the integral exists;
$\displaystyle A(x) = \int_a^x f(t)\,dt$	a function of x;
$\displaystyle\int f(x)\,dx$	a family of antiderivatives.

For continuous f, FTC relates these: $A' = f$, and $\int_a^b f(x)\,dx = F(b) - F(a)$ for any antiderivative satisfying the endpoint hypotheses.

Basic formulas. Each identity follows by differentiation on the stated domain:

$$\int x^p\,dx = \frac{x^{p+1}}{p+1} + C \qquad (x > 0,\ p \neq -1),$$

$$\int \frac{1}{x}\,dx = \ln|x| + C \qquad \text{on an interval avoiding } 0,$$

$$\int e^x\,dx = e^x + C \qquad (x \in \mathbb{R}),$$

$$\int \cos x\,dx = \sin x + C, \qquad \int \sin x\,dx = -\cos x + C \qquad (x \in \mathbb{R}),$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + C \qquad (x \in \mathbb{R}).$$

For nonnegative integer p, the power formula holds on all of $\mathbb{R}$. Other extensions require checking the domain. If $F' = f$ and $G' = g$, then $(\alpha F + \beta G)' = \alpha f + \beta g$, so antiderivatives of sums and scalar multiples are obtained term by term.

Example

On $(0, \infty)$,

$$\int \left(3x^2 - \frac{2}{x}\right) dx = x^3 - 2\ln x + C, \qquad \int_1^e \left(3x^2 - \frac{2}{x}\right) dx = e^3 - 3.$$

The constant cancels in the definite integral's endpoint difference.

2. Net change and distance

Net-change theorem. Suppose Q is continuous on $[a, b]$, differentiable on (a, b), and $Q' = r$ there, where r is continuous on $[a, b]$. FTC II gives

$$\boxed{Q(b) - Q(a) = \int_a^b r(t)\,dt.}$$

Thus, for a particle with position $x(t)$ and continuous velocity $v(t) = x'(t)$,

$$x(T) - x(0) = \int_0^T v(t)\,dt.$$

This proves the connection between velocity and change in position that motivated our definition of the integral.

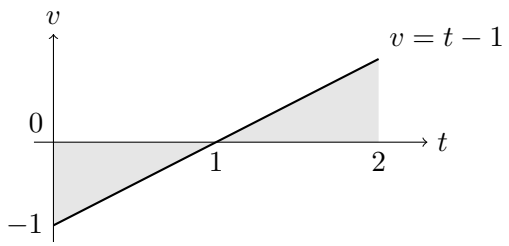
Distance. Where $v \geq 0$, position is nondecreasing and distance equals displacement. Where $v \leq 0$, distance is the negative of displacement. For finitely many changes of direction, split the time interval at those changes and add the distances. This gives

$$\boxed{\text{total distance on } [a, b] = \int_a^b |v(t)|\,dt.}$$

The same formula holds more generally for continuous velocity; the finite-splitting argument proves it when there are finitely many changes of direction.

Example

Let $v(t) = t - 1$ for $0 \leq t \leq 2$. The particle moves in the negative direction until $t = 1$, then in the positive direction.



Its displacement is

$$\int_0^2 (t - 1)\,dt = \left[\frac{t^2}{2} - t\right]_0^2 = 0,$$

whereas its total distance is

$$\int_0^1 (1 - t)\,dt + \int_1^2 (t - 1)\,dt = \frac{1}{2} + \frac{1}{2} = 1.$$

The graph has two triangular regions of equal area. Their contributions cancel in the signed integral of velocity. The integral of speed counts both areas positively. The particle returns to its starting position.

3. Substitution in definite integrals

Theorem. Let g be continuously differentiable on $[a, b]$, with $a < b$, and let f be continuous on an open interval J containing $g([a, b])$. Then

$$\boxed{\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.}$$

Here continuous differentiability means that g is continuous on $[a, b]$, differentiable inside, and its derivative extends continuously to the endpoints. The integral on the right uses our convention for reversed or equal bounds.

Proof. Choose F with $F' = f$ on J, as guaranteed by FTC I. The chain rule shows that $F \circ g$ is an antiderivative of $f(g(x))g'(x)$. The integrand is continuous, so FTC II gives

$$\int_a^b f(g(x))g'(x)\,dx = F(g(b)) - F(g(a)).$$

Applying FTC II to f, with orientation if necessary, gives the same endpoint difference for $\int_{g(a)}^{g(b)} f(u)\,du$. This proves the formula. □

Substitution notation. Write $u = g(x)$ and $du = g'(x)\,dx$. This records the derivative factor in the chain rule used in the proof. The new bounds are the values of u at the original endpoints. With these bounds, the calculation can remain in u throughout. Alternatively, first find an antiderivative in x and use the original bounds. In either calculation, the variable and its bounds must agree.

Example

Set $u = 1 + x^2$, $du = 2x\,dx$. At $x = 0$, $u = 1$; at $x = 1$, $u = 2$. Hence

$$\int_0^1 2x\sqrt{1 + x^2}\,dx = \int_1^2 u^{1/2}\,du = \left[\frac{2}{3}u^{3/2}\right]_1^2 = \frac{2}{3}(2\sqrt{2} - 1).$$

Example

Set $u = 1 - x$, $du = -dx$. The bounds become 1 and 0:

$$\int_0^1 e^{1-x}\,dx = -\int_1^0 e^u\,du = \int_0^1 e^u\,du = e - 1.$$

The minus sign comes from the derivative; the order of the bounds comes from the endpoint values. Both must be included.

Monotonicity of g is not required: the proof uses only the endpoint values of $F \circ g$. The equal-bounds example in Section 7 illustrates this.

4. Substitution in indefinite integrals

Let f be continuous on an open interval J, and choose an antiderivative F there. Let g be differentiable on an open interval I, with $g(I) \subseteq J$. The chain rule gives

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

Consequently,

$$\boxed{\int f(g(x))g'(x)\,dx = F(g(x)) + C, \qquad F' = f.}$$

The Mean Value Theorem ensures that this gives every antiderivative on I, up to a constant.

Calculation. We write $u = g(x)$ and $du = g'(x)\,dx$. Then the calculation becomes

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du = F(u) + C = F(g(x)) + C.$$

Here $du = g'(x)\,dx$ records the derivative factor in the chain rule. The displayed equality of indefinite integrals is understood with $u = g(x)$. The final answer to the original problem is a function of x. No division by $g'(x)$ is required, so its zeros do not invalidate the formula.

Example

Find $\int 2x\cos(x^2)\,dx$. Set $u = x^2$, so $du = 2x\,dx$. Then

$$\int 2x\cos(x^2)\,dx = \int \cos u\,du = \sin u + C = \sin(x^2) + C.$$

Differentiation checks the answer: $(\sin(x^2))' = 2x\cos(x^2)$. It is valid on all of $\mathbb{R}$, including at $x = 0$.

Example

Find $\int x(1+x^2)^4\,dx$. With $u = 1 + x^2$, $du = 2x\,dx$, so $x\,dx = \frac{1}{2}\,du$ in the substitution:

$$\int x(1+x^2)^4\,dx = \frac{1}{2}\int u^4\,du = \frac{u^5}{10} + C = \frac{(1+x^2)^5}{10} + C.$$

Indeed, the derivative of the final expression is $\frac{1}{10}\cdot 5(1+x^2)^4\cdot 2x = x(1+x^2)^4$.

5. Logarithmic antiderivatives

For $u > 0$, $(\ln|u|)' = (\ln u)' = 1/u$. For $u < 0$, $\ln|u| = \ln(-u)$, and the chain rule gives

$$\frac{d}{du}\ln(-u) = \frac{-1}{-u} = \frac{1}{u}.$$

Thus $\ln|u|$ is an antiderivative of $1/u$ on each interval avoiding zero. If g is differentiable and nonzero on an open interval I, another application of the chain rule gives

$$\boxed{\int \frac{g'(x)}{g(x)}\,dx = \ln|g(x)| + C \quad \text{on } I.}$$

The absolute value allows a negative value of g; it does not allow $g(x) = 0$. Constants on different intervals of the domain can be different.

Example

With $u = x^2 - 1$, $du = 2x\,dx$, we obtain

$$\int \frac{2x}{x^2-1}\,dx = \ln|x^2 - 1| + C.$$

This holds separately on $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$. On the middle interval, the expression is $\ln(1 - x^2) + C$; its derivative is again $2x/(x^2 - 1)$. For example,

$$\int_0^{1/2} \frac{2x}{x^2-1}\,dx = [\ln(1-x^2)]_0^{1/2} = \ln\frac{3}{4}.$$

The negative answer agrees with the sign of the integrand.

The same expression does not evaluate a Riemann integral across $x = 1$ or $x = -1$. The integrand is undefined and unbounded near those points. For example, $\int_0^2 2x/(x^2-1)\,dx$ is not a Riemann integral on $[0, 2]$; subtracting the endpoint values of $\ln|x^2 - 1|$ would not be a valid application of FTC.

Example

For $\displaystyle\int_1^e \frac{(\ln x)^2}{x}\,dx$, set $u = \ln x$, so $du = dx/x$. The original interval lies in $(0, \infty)$, and its endpoints become 0 and 1:

$$\int_1^e \frac{(\ln x)^2}{x}\,dx = \int_0^1 u^2\,du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}.$$

6. Choosing a substitution

A substitution should express the entire integrand and differential in the new variable. Sometimes this requires rewriting a remaining factor. The candidate antiderivative can always be checked by differentiation.

Example

Find $\int x\sqrt{x+1}\,dx$ on $(-1,\infty)$. Choose $u = x+1$. Then $du = dx$ and $x = u-1$, giving

$$\begin{aligned}\int x\sqrt{x+1}\,dx &= \int (u-1)u^{1/2}\,du \\ &= \int (u^{3/2} - u^{1/2})\,du \\ &= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C \\ &= \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + C.\end{aligned}$$

Checking the derivative gives

$$(x+1)^{3/2} - (x+1)^{1/2} = \sqrt{x+1}\big((x+1)-1\big) = x\sqrt{x+1}.$$

Example

Find $\int 1/(9+x^2)\,dx$. To obtain the known derivative $1/(1+u^2)$, set $u = x/3$, so $x = 3u$, $dx = 3\,du$. Then

$$\begin{aligned}\int \frac{1}{9+x^2}\,dx &= \int \frac{3}{9+9u^2}\,du = \frac{1}{3}\int \frac{1}{1+u^2}\,du \\ &= \frac{1}{3}\arctan u + C = \frac{1}{3}\arctan\frac{x}{3} + C.\end{aligned}$$

The formula holds for every real x. Its derivative is

$$\frac{1}{3}\,\frac{1}{1+(x/3)^2}\,\frac{1}{3} = \frac{1}{9+x^2}.$$

7. Further examples

Example

For $\int \sin x \cos x \, dx$, taking $u = \sin x$ gives

$$\int \sin x \cos x \, dx = \int u \, du = \frac{1}{2} \sin^2 x + C.$$

Taking $u = \cos x$ instead gives

$$\int \sin x \cos x \, dx = -\int u \, du = -\frac{1}{2} \cos^2 x + C.$$

The displayed antiderivatives differ by $1/2$, since $\sin^2 x + \cos^2 x = 1$. They therefore describe the same family when the arbitrary constant is included. In particular,

$$\int_0^{\pi/2} \sin x \cos x \, dx = \frac{1}{2}.$$

Example

Set $g(x) = x^2$ on $[-1, 1]$, and $f(u) = e^u$. The hypotheses of definite substitution hold even though g is not monotone on this interval. Its endpoint values are equal, so

$$\int_{-1}^{1} 2xe^{x^2} \, dx = \int_1^1 e^u \, du = 0.$$

This is cancellation of signed contributions. Splitting at zero makes it explicit:

$$\int_{-1}^{0} 2xe^{x^2} \, dx = \int_1^0 e^u \, du = 1 - e, \qquad \int_0^1 2xe^{x^2} \, dx = \int_0^1 e^u \, du = e - 1.$$

The corresponding integral of the absolute value is

$$\int_{-1}^{1} \left|2xe^{x^2}\right| \, dx = 2(e - 1).$$

For the absolute value, we must account for the sign of $2x$ on each half. Equal transformed endpoints do not make that integral zero.

8. Integration by parts for definite integrals

Theorem. Let u and v be continuously differentiable on $[a, b]$, with $a < b$. Then

$$\boxed{\int_a^b u(x)v'(x)\,dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,dx.}$$

Here $[u(x)v(x)]_a^b = u(b)v(b) - u(a)v(a)$. All integrands are continuous on the interval, so both integrals exist.

Proof. The product rule gives

$$(uv)'(x) = u'(x)v(x) + u(x)v'(x).$$

Integrate over $[a, b]$. The Fundamental Theorem and linearity give

$$u(b)v(b) - u(a)v(a) = \int_a^b u'(x)v(x)\,dx + \int_a^b u(x)v'(x)\,dx.$$

Rearranging proves the formula. □

Integration by parts replaces the integral of uv' by an endpoint term and the integral of $u'v$. To use it, choose a factor to serve as u, differentiate it, and find an antiderivative v of the remaining factor v'. The new integral should be easier to evaluate.

Example

Evaluate $\int_0^1 xe^x\,dx$. Choose

$$u(x) = x, \qquad v'(x) = e^x, \qquad u'(x) = 1, \qquad v(x) = e^x.$$

The theorem gives

$$\begin{aligned}\int_0^1 xe^x\,dx &= [xe^x]_0^1 - \int_0^1 e^x\,dx \\ &= e - [e^x]_0^1 \\ &= e - (e - 1) = 1.\end{aligned}$$

The same bounds remain on the new integral. In the boundary term, we evaluate the product at each endpoint and subtract.

Substitution follows from the chain rule; integration by parts follows from the product rule. The Fundamental Theorem turns each derivative identity into a formula for definite integrals.